TPG4190 Seismic data acquisition and processing Lecture 18: Multiples - Surface Related Multiple Removal (SRME)

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Surface Related Multiple Removal (SRME)

- Multiple reflections generated by the sea surface are usually strong
- Mathematical removal of the sea surface removes surface multiples
- SRME does not rely on moveout discrimination

Model for multiple reflections

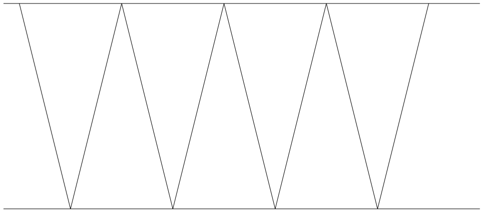

Figure: Raypaths for multiple reflections

Model for multiple reflections

- Vertically traveling waves
- τ: Traveltime
- r: Reflection coefficient at the bottom
- Reflection coefficient at the top is equal to -1

$$p(t) = rs(t-\tau) - r^2 s(t-2\tau) + r^3 s(t-3\tau) + \dots \tag{1}$$

Model for multiple reflections

Taking the Fourier transform of equation (1), one gets

$$P(\omega) = rS(\omega)\exp(-i\omega\tau) - r^2 S(f)\exp(-2i\omega\tau) + r^3 \exp(-3i\omega\tau) + \dots$$

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If we set $R = r\exp(-i\omega\tau)$, then equation (2) is:

$$P(\omega) = S(\omega)R\left[1 - R + R^2 + \dots\right]. \tag{3}$$

Model for multiple reflections

The right hand side of equation (1) is an infinite geometrical series,

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \dots \tag{4}$$

implying

$$P(\omega) = \frac{S(\omega)R}{1+R}. \tag{5}$$

Removal of multiples

Equation (5) solved with respect to R

$$R = \frac{P(\omega)}{S(\omega)[1 - S^{-1}(f)P(\omega)]}, \tag{6}$$

or

$$S(\omega)R = \frac{P(\omega)}{[1 - S^{-1}(f)P(\omega)]}. \tag{7}$$

Use of (Rottmann (1960))

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \ldots, \tag{8}$$

gives

Removal of multiples

$$P_0(f) = P(\omega)\left[1 + S^{-1}P(\omega) + S^{-}2P^2(\omega) + \ldots\right], \tag{9}$$

where $P_0(\omega) = S(\omega)R(\omega)$ is the primary response,

Removal of multiples

- Remove multiple reflections by summing powers of the recorded data itself
- (9) does *not* require any knowledge of the water depth nor the reflection coefficient!

Removal of multiples

To see how the summation works, truncate the series

$$P_0(\omega) \approx P(\omega) + S^{-1}P^2(\omega), \tag{10}$$

and insert equation (5) into equation (10)

$$P_0(\omega) \approx \frac{S(\omega)R(\omega)}{1+R(\omega)} + S^{-1}(\omega)\frac{S^2(\omega)R^2(\omega)}{[1+R(\omega)]^2}. \tag{11}$$

Using the following relations

$$\begin{aligned} \frac{1}{1+x} &= 1 - x + \ldots, \\ \frac{1}{(1+x)^2} &= 1 - 2x + \ldots, \end{aligned} \tag{12}$$

Removal of multiples

we have

$$\frac{S(\omega)R(\omega)}{1+R(\omega)} \approx S(\omega)R(\omega)\left[1-R(\omega)\right], \tag{13}$$

and

$$S^{-1}(\omega)\frac{S^2(f)R^2(\omega)}{[1+R(\omega)]^2} \approx S^{-1}(\omega)S^2(\omega)R^2(\omega)[1-2R(\omega)], \tag{14}$$

equation (11) becomes

$$P_0(\omega) \approx S(\omega)R(\omega)\left[1-R(\omega)\right] + S^{-1}(\omega)S^2(\omega)R^2(\omega)(1-2R(\omega)), \tag{15}$$

which is equal to

$$P_0(\omega) \approx S(\omega)R(\omega) - S(\omega)R^2(\omega) + S(\omega)R^2(\omega) - 2S(\omega)R^3(\omega), \tag{16}$$

which is equal to

$$P_0(\omega) \approx S(\omega)R(\omega) - 2S(\omega)R^3(\omega). \tag{17}$$

Removal of multiples

- *first order* multiples have been removed
- Third order and higher multiples remain.
- Third order and higher multiples remain.
- Keeping more terms in equation (10) *second* and higher order would be removed
- Method also works for non-vertical waves
- Method also works for more than on layer

SRME 3 D

From reciprocity we get the integral representation in the frequency domain

$$\begin{aligned} P(\boldsymbol{x}_r, \boldsymbol{x}_s) &= S(\omega) G(\boldsymbol{x}_r, \boldsymbol{x}_s) \\ &+ \int dS(\boldsymbol{x}') \left[G(\boldsymbol{x}_r, \boldsymbol{x}') \nabla P(\boldsymbol{x}', \boldsymbol{x}_s) - \nabla G(\boldsymbol{x}_r, \boldsymbol{x}') P(\boldsymbol{x}', \boldsymbol{x}_s) \right] \cdot \boldsymbol{n} \end{aligned}$$

Here

- G: Greens function (impulse response) with transparent boundaries
- P: Pressure with boundary conditions
- S: Source pulse

SRME 3D

Set $P_0 = S(\omega)G$

$$\begin{aligned} P(\boldsymbol{x}_r, \boldsymbol{x}_s) &= P_0(\boldsymbol{x}_r, \boldsymbol{x}_s) \\ &+ S(\omega)^{-1} \int dS(\boldsymbol{x}') \left[P_0(\boldsymbol{x}_r, \boldsymbol{x}') \nabla P(\boldsymbol{x}', \boldsymbol{x}_s) \right. \\ &- \left. \nabla P_0(\boldsymbol{x}_r, \boldsymbol{x}') P(\boldsymbol{x}', \boldsymbol{x}_s) \right] \cdot \boldsymbol{n}(\boldsymbol{x}) \end{aligned}$$

Solve for P_0

$$\begin{aligned} P_0(\boldsymbol{x}_r, \boldsymbol{x}_s) &= P(\boldsymbol{x}_r, \boldsymbol{x}_s) \\ &- S(\omega)^{-1} \int dS(\boldsymbol{x}') \left[P_0(\boldsymbol{x}_r, \boldsymbol{x}') \nabla P(\boldsymbol{x}', \boldsymbol{x}_s) \right. \\ &- \left. \nabla P_0(\boldsymbol{x}_r, \boldsymbol{x}') P(\boldsymbol{x}', \boldsymbol{x}_s) \right] \cdot \boldsymbol{n}(\boldsymbol{x}) \end{aligned}$$

SRME 3D

Simplification: $\nabla P \cdot \boldsymbol{n} \approx P$ Solve for P_0

$$\begin{aligned} P_0(\boldsymbol{x}_r, \boldsymbol{x}_s) &= P(\boldsymbol{x}_r, \boldsymbol{x}_s) \\ &- S(\omega)^{-1} \int dS(\boldsymbol{x}') P_0(\boldsymbol{x}_r, \boldsymbol{x}') P(\boldsymbol{x}', \boldsymbol{x}_s). \end{aligned} \tag{18}$$

This is an integral equation for the data without multiples in terms of data with multiples and can be solved by recursion.

SRME 3D

Set $P = P_0$ on the right hand side of equation (18) to get

$$\begin{aligned} P_0^{(1)}(\boldsymbol{x}_r, \boldsymbol{x}_s) &= P(\boldsymbol{x}_r, \boldsymbol{x}_s) \\ &- S(\omega)^{-1} \int dS(\boldsymbol{x}') P(\boldsymbol{x}_r, \boldsymbol{x}') P(\boldsymbol{x}', \boldsymbol{x}_s). \end{aligned} \tag{19}$$

This equation is similar to the one-dimensional result (9), with the addition of a spatial integration. To get the next term insert equation (19) into equation (18)

$$\begin{aligned} P_0^{(2)}(\boldsymbol{x}_r, \boldsymbol{x}_s) &= P(\boldsymbol{x}_r, \boldsymbol{x}_s) \\ &- S(\omega)^{-1} \int dS(\boldsymbol{x}') P_0^{(1)}(\boldsymbol{x}_r, \boldsymbol{x}') P(\boldsymbol{x}', \boldsymbol{x}_s). \end{aligned} \tag{20}$$

SRME 3D

Written out this is:

$$P_0^{(2)}(\boldsymbol{x}_r, \boldsymbol{x}_s) = P(\boldsymbol{x}_r, \boldsymbol{x}_s)$$

$$-S^{-1} \int dS(\boldsymbol{x}') \left[P(\boldsymbol{x}_r, \boldsymbol{x}') - S^{-1} \int dS(\boldsymbol{x}'') P(\boldsymbol{x}_r, \boldsymbol{x}'') P(\boldsymbol{x}'', \boldsymbol{x}') \right] P(\boldsymbol{x}', \boldsymbol{x}_s)$$

SRME 3D

$$P_0^{(2)}(\boldsymbol{x}_r, \boldsymbol{x}_s) = P(\boldsymbol{x}_r, \boldsymbol{x}_s) - S^{-1} \int dS(\boldsymbol{x}') P(\boldsymbol{x}_r, \boldsymbol{x}') P(\boldsymbol{x}', \boldsymbol{x}_s) \quad (22)$$

$$+ S^{-2} \int dS(\boldsymbol{x}') \int dS(\boldsymbol{x}'') P(\boldsymbol{x}_r, \boldsymbol{x}'') P(\boldsymbol{x}'', \boldsymbol{x}') P(\boldsymbol{x}', \boldsymbol{x}_s) \quad (23)$$

This is again the same structure as the one-dimensional case.

SRME 3D

$$\begin{aligned} P_0^{(i+1)}(\boldsymbol{x}_r, \boldsymbol{x}_s) &= P(\boldsymbol{x}_r, \boldsymbol{x}_s) \\ &- S(\omega)^{-1} \int dS(\boldsymbol{x}') P_0^{(i)}(\boldsymbol{x}_r, \boldsymbol{x}') P(\boldsymbol{x}', \boldsymbol{x}_s). \quad (24) \end{aligned}$$

Where $P_0^1 = P$.

$$\begin{aligned} P_0^{(i+1)}(\boldsymbol{x}_r, \boldsymbol{x}_s) &= P(\boldsymbol{x}_r, \boldsymbol{x}_s) \\ &- S(\omega)^{-1} \sum_x \Delta x \sum_y \Delta y P_0^{(i)}(\boldsymbol{x}_r; x, y) P(x, y; \boldsymbol{x}_s). \end{aligned}$$

3D SRME Example

Hokstad and Sollie, 2006, Geophysics.

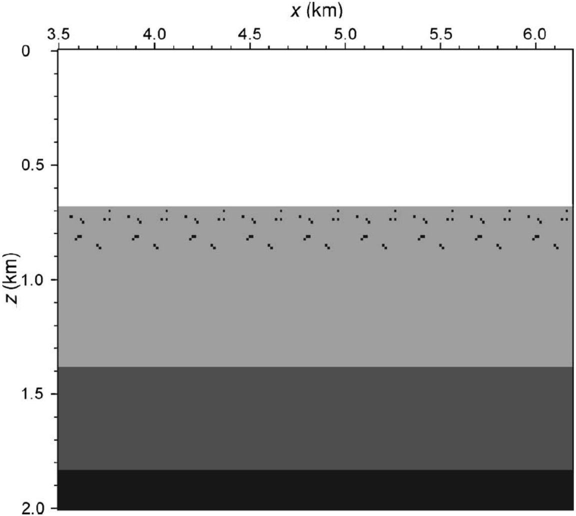
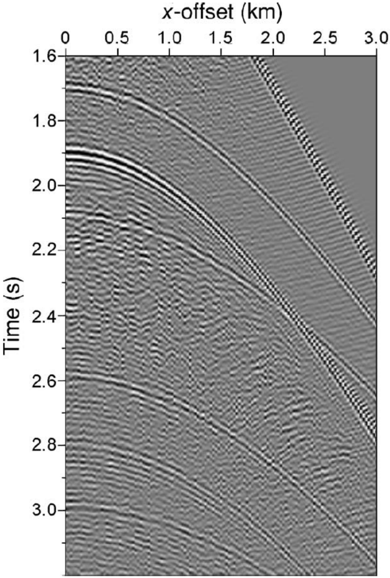


Figure 1. Vertical slice from the 3D diffractor model. The model has discrete translation invariance in both x- and y-directions.

3D SRME Example



3D SRME Example

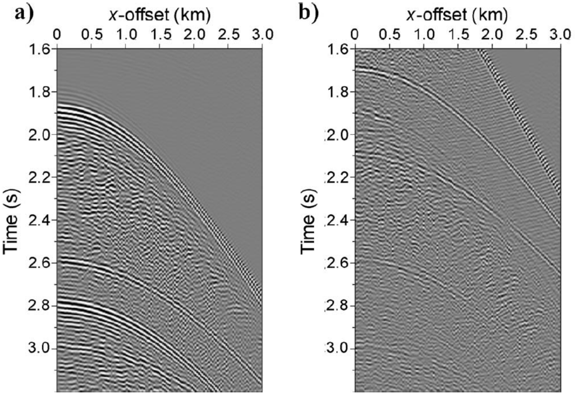


Figure 4. The 2D SRME result. (a) Predicted multiples and (b) after adaptive subtraction.

3D SRME Example

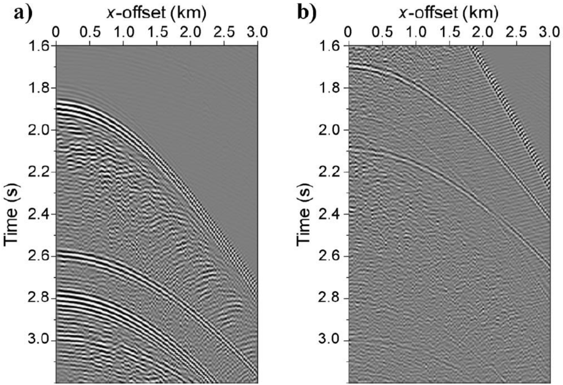
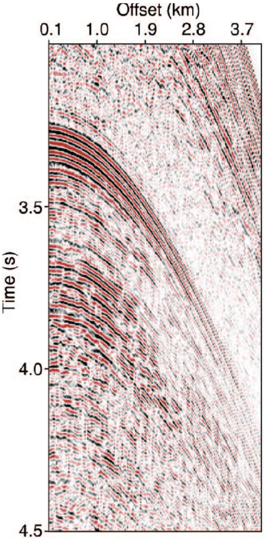
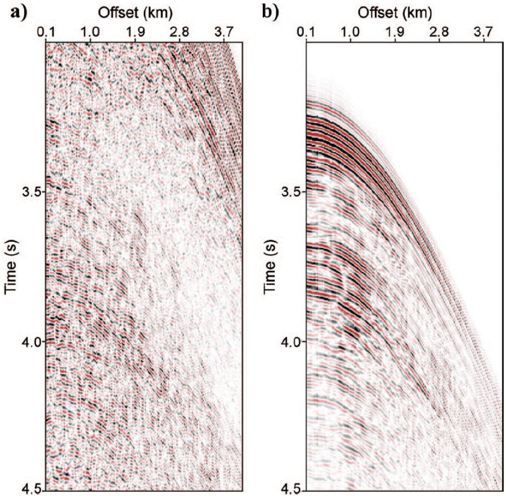


Figure 5. The 3D SRME reference with dense source and receiver sampling. (a) Predicted multiples and (b) after adaptive subtraction.

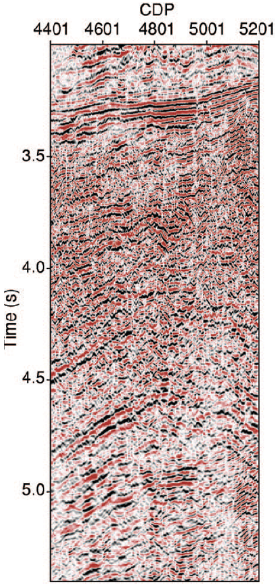
3D SRME Example



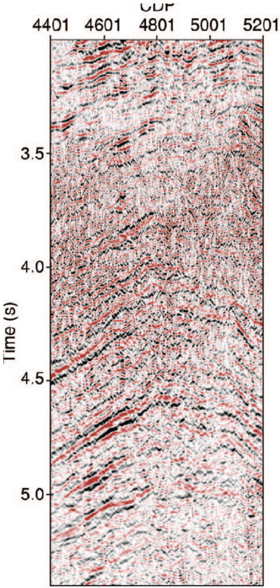
3D SRME Example



3D SRME Example



3D SRME Example



SRME

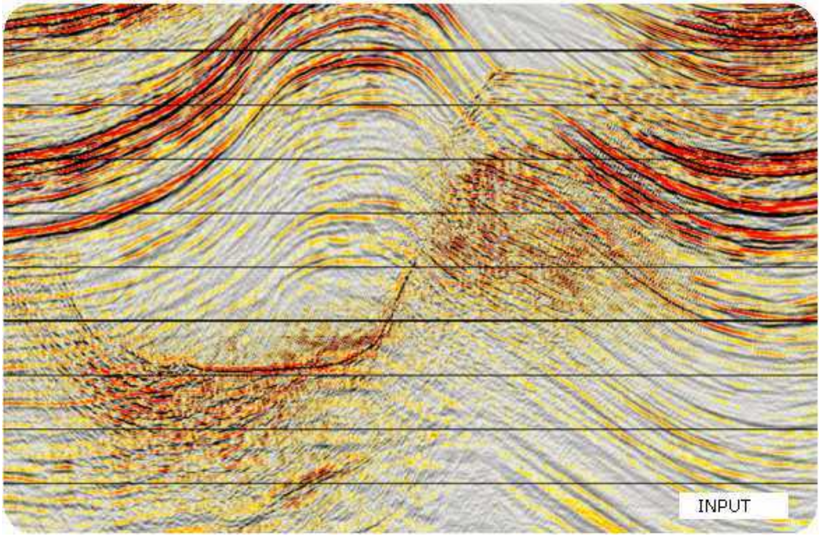


Figure: Data with multiples (Courtesy of CGG)

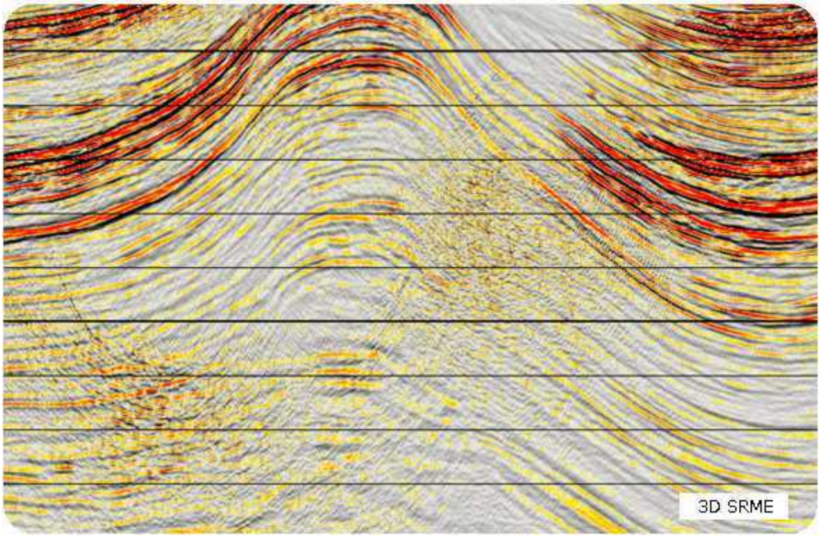


Figure: Multiples removed using SRME (Courtesy of CGG)